

Relation Algebra

§11. Relator

(11a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

(11b)

the statement

$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$.

Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.

$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.

Two relators agreeing on maps are equal.

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$

$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$

The **power relator** $P \multimap x \sqsubseteq P(R) \ y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§11.1. Fork $\langle R, S \rangle$

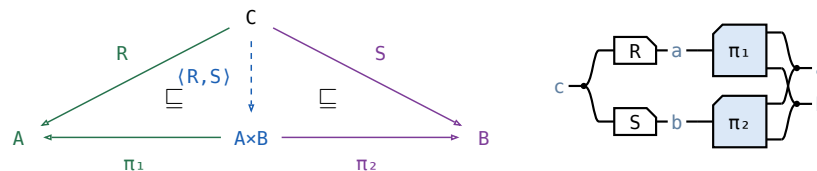
(11.1a)

The **fork** of $R : \mathcal{C} \rightarrow \mathcal{A}$ and $S : \mathcal{C} \rightarrow \mathcal{B}$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

(11.1b)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad \langle R, S \rangle \pi_2 = \text{dom}(R) S$$

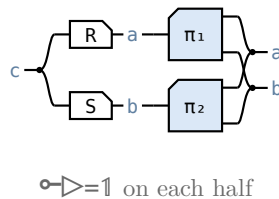
(11.1c)



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In $\mathbf{Rel}$, $c \sqsubseteq \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = 1 \otimes \circ$ discards the second component, so $\pi_1^\circ = 1 \otimes \bullet$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(11.1d)

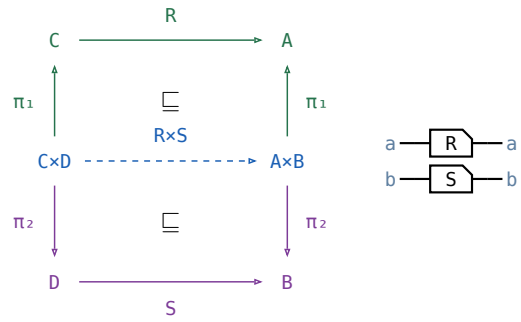


§11.1.1. Relational product $R \times S$

(11.1.1a)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

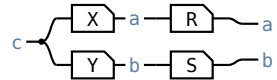
(11.1.1b)



Right-then-up is $(R \times S)\pi_1$, up-then-right is $\pi_1 R$, and $(R \times S)\pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§11.1.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



(11.1.2a)

(11.1.2b)

the law	picture
1 $R \times S \triangleq (\pi_1 R, \pi_2 S)$	
2 $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ both sides are the same strokes — absorption. Its $\sqsubseteq$ half is that half at $S := \mathbb{1}$ and at $R := \mathbb{1}$ — stages of their proof of this row; the corollary is the $(R \times S) (U \times V) = (RU) \times (SV)$ it yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$.	
3 $\langle R, S \rangle \pi_1 = \text{dom}(S) R$ (11.1b)	
4 $\langle R, S \rangle \pi_2 = \text{dom}(R) S$ (11.1b)	
5 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$	
6 $\langle R, S \rangle^\circ (U, V) \sqsubseteq (R^\circ U) \times (S^\circ V)$	
7 $\text{Map}(f) \implies f \langle R, S \rangle = \langle fR, fS \rangle$ f a map; it fails for an arbitrary arrow;	
8 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$; only $\sqsubseteq$ for an arbitrary R, S	
9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§11.2. Coproduct $[R, S] : A + B \rightarrow C$

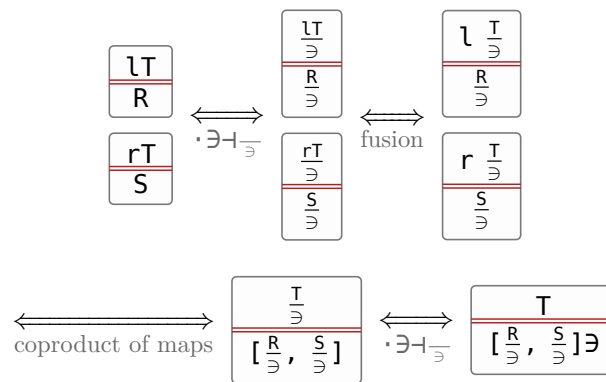
(11.2a)

the statement	picture
$[R, S] \triangleq \mathfrak{l} \circ R \cup r \circ S$ The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
$R + S \triangleq [R\mathfrak{l}, S\mathfrak{r}]$	
$\mathfrak{l}[R, S] = R, r[R, S] = S$ $\mathfrak{l}T = R \wedge rT = S \implies T = [R, S]$	
$\mathfrak{l}\mathfrak{l}^\circ = 1, r\mathfrak{r}^\circ = 1$	
$\mathfrak{l}r^\circ = \emptyset, r\mathfrak{l}^\circ = \emptyset$	
$\mathfrak{l}^\circ \mathfrak{l} \cup r^\circ r = 1$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$	

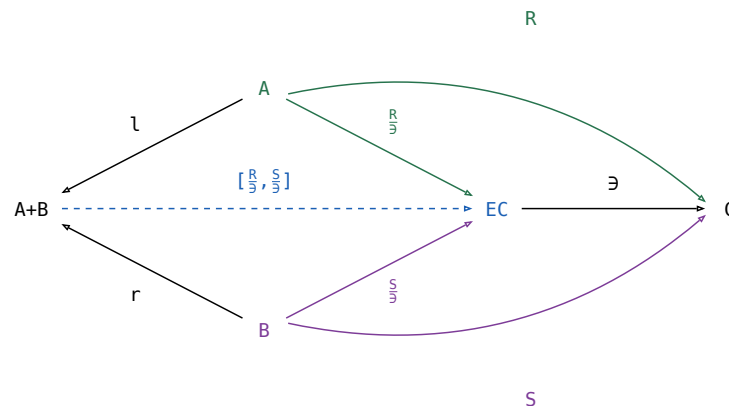
§11.2.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free: $\mathfrak{l}, r$ were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow EC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(11.2.1a)



(11.2.1b)



Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists = l \circ R \cup r \circ S$.

(11.2.1c)

the law	picture
$[R, S] \triangleq l \circ R \cup r \circ S$ (11.2a)'s first row	
$R + S \triangleq [Rl, Sr]$	
$[U, V] \circ [R, S] = U \circ R \cup V \circ S$ (11.2a)'s last row;	
$[1, 0] = l \circ, [0, 1] = r \circ$ $[l, r] = 1$ which is (5.9)	
prove (5.11), and say why duality does not carry it over from the product law	
$(R + S) \cap [P, Q] [U, V] \circ =$ $(R \cap P U \circ) + (S \cap Q V \circ)$ $[P, Q] [U, V] \circ$ is a full 2×2 of composites; $R + S$ is diagonal, so the meet cuts the two off-diagonal branches	

§11.3. The power relator $P(R)$

(11.3a)

For $R : A \rightarrow B$,

$$P(R) \triangleq (\epsilon \setminus R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

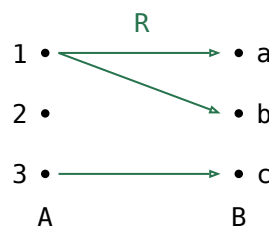
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(11.3b)

	in Rel	in words	$\text{im}(xs)$	partner
$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	every y with some $xs \ R \ y$ is in ys	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{ every } y \text{ with some } xs \ R \ y$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall x. x \ R \ \text{some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$	—	every x and every y has a partner

(11.3c)



xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

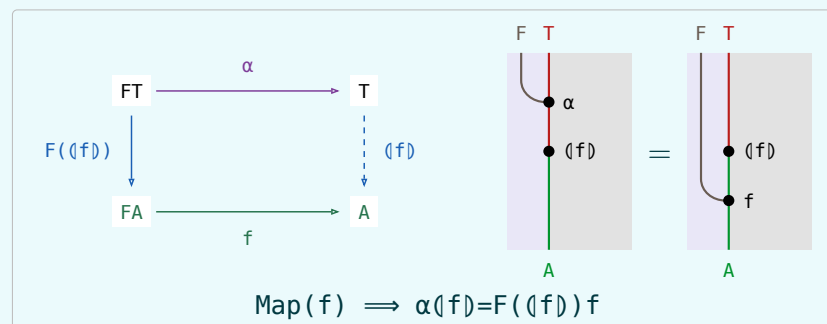
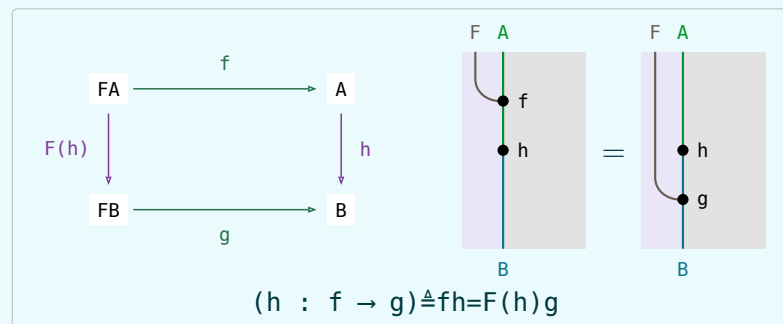
(11.3d)

law	the reading
$X \sqsubseteq P(R) \iff X \sqsubseteq \exists R \text{ and } X^\circ \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(RS) = P(R)P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§11.4. Initial algebra

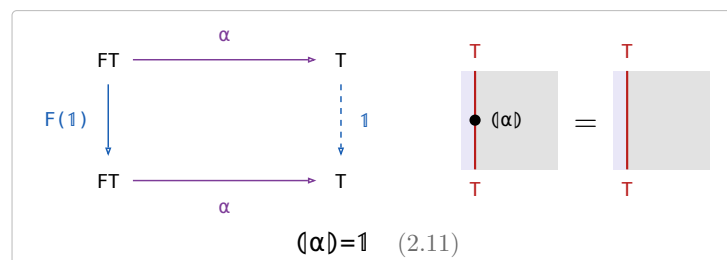
(11.4a)

An **F-algebra** is a map $f : F(A) \rightarrow A$; A is its **carrier**. An **F-homomorphism** from $f : F(A) \rightarrow A$ to $g : F(B) \rightarrow B$ is a map $h : A \rightarrow B$ with $f \circ h = F(h) \circ g$. The **initial algebra** $\alpha : F(T) \rightarrow T$ is the F-algebra with exactly one F-homomorphism $\langle f \rangle : T \rightarrow A$ to every F-algebra f .



§11.4.1. Reflection

(11.4.1a)

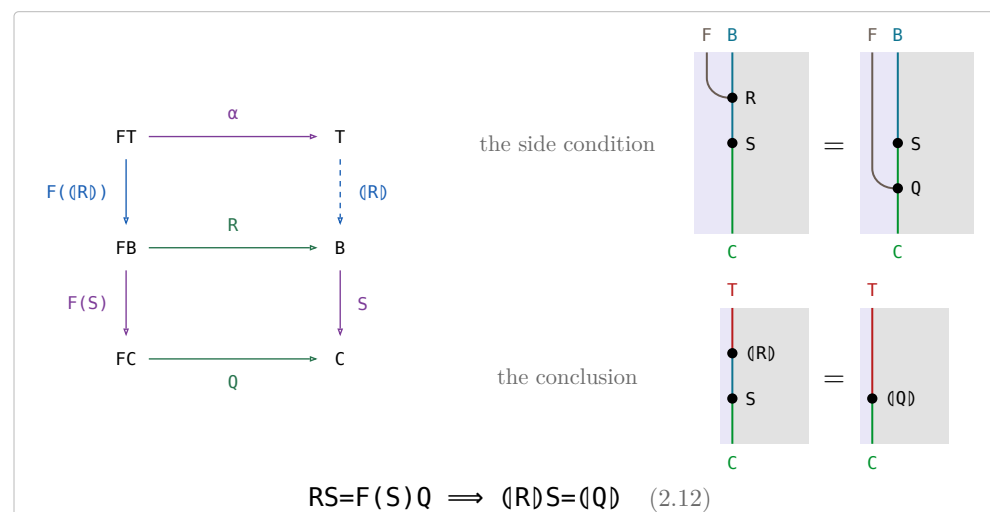


Taking a value apart with α and putting it straight back is doing nothing.

§11.4.2. Fusion: $(R:FB)$ absorb $S:R \rightarrow Q$ (in $\text{Alg}(F)$) and becomes $\langle Q \rangle$

When $S : B \rightarrow C$ is an F-homomorphism from $R : F(B) \rightarrow B$ to $Q : F(C) \rightarrow C$, folding with R and then applying S is folding with Q .

(11.4.2a)



§11.5. Type relator

(11.5a)

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha)$$

(11.5b)

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way.
functor	$T(1) = 1$ and $T(R)T(S) = T(RS)$	Acting by the identity changes nothing, and two actions in a row are one action.
type functor fusion	$T(f)(h) = (F(f, 1)h)$	A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because F is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$.
naturality of α	$\alpha T(f) = F(f, T(f)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

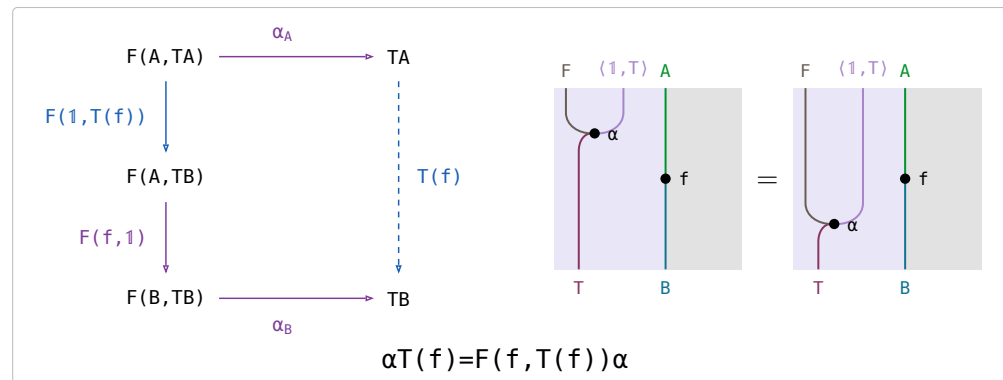
§11.5.1. Type functor

(11.5.1a)

Let F be a bifunctor taking both the parameter A and the recursive position TA , with an initial algebra $\alpha_A : F(A, TA) \rightarrow TA$ for every object A . Then T is a functor, acting on a map $f : A \rightarrow B$ by

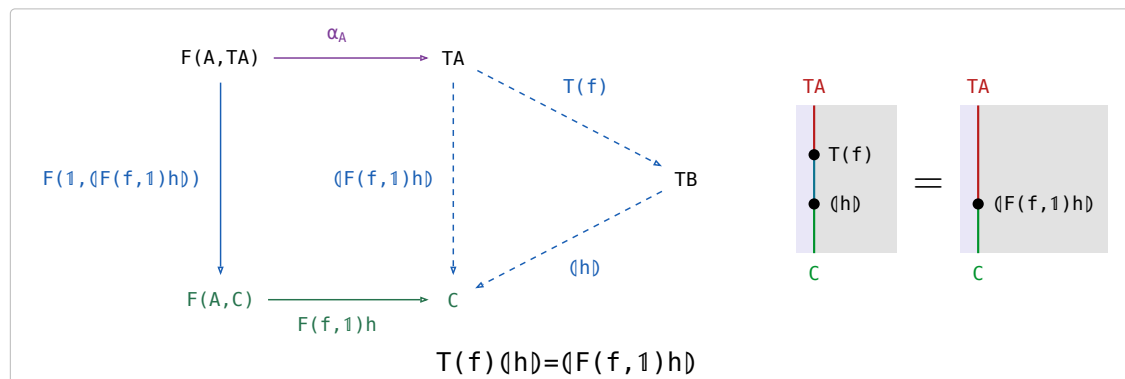
$$T(R) \triangleq (F(R, 1) \alpha)$$

(11.5.1b)



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ is a bifunctor and a wire is a unary functor, so the two arguments are packed first: $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$ sends A to (A, TA) , and $F(\langle 1, T \rangle(A))$ is $F(A, TA)$.
- The picture is three wires — F , $\langle 1, T \rangle$, and the object — and the region between the first two is $\mathcal{A} \times \mathcal{A}$.
- α is then an ordinary natural transformation $F \circ \langle 1, T \rangle \Rightarrow T$: its bead eats the F and $\langle 1, T \rangle$ wires, and the T wire is born under it.
- $F(f, T(f))$ costs no notation. It is the bead f on the object wire with $\langle 1, T \rangle$ and F running past: $\langle 1, T \rangle$ is what turns f into the pair $(f, T(f))$, and F is what applies it.
- The law is the naturality of α , which is exactly the freedom to slide that f bead past it.

(11.5.1c)



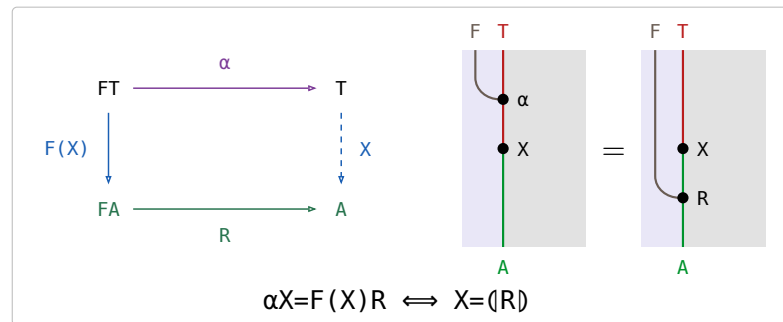
§11.6. Reduce

(11.6a)

let F be a relator and has **initial algebra** $\alpha : F(T) \rightarrow T$ in the subcategory of functions. α is also initial in the allegory:

§11.6.1. The defining equation

(11.6.1a)



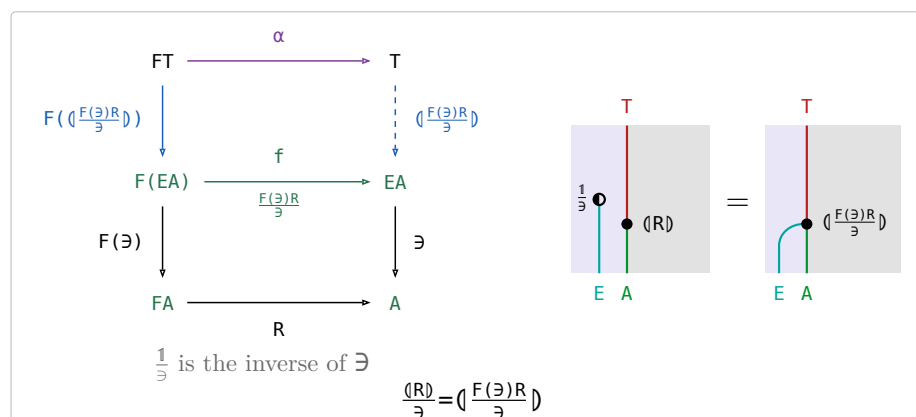
An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and A .

(11.6.1b)

$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$		
catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
part	Nat	[A]
datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
the fold, pointwise	$\llbracket [c, f] \rrbracket (\text{zero}) = c$ $\llbracket [c, f] \rrbracket (\text{succ}(n)) = f(\llbracket [c, f] \rrbracket (n))$	$\llbracket [c, f] \rrbracket (\text{nil}) = c$ $\llbracket [c, f] \rrbracket (\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket (x))$

§11.6.2. $\llbracket (R) \rrbracket = (\llbracket \frac{F(\exists)R}{\exists} \rrbracket) \exists$

(11.6.2a)



(11.6.2b)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \iff \boxed{\frac{\alpha X}{\frac{F(X)R}{\exists}}} \iff \boxed{\frac{\alpha X}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \iff \boxed{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}} \iff \boxed{\frac{\frac{X}{\exists}}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket}} \iff \boxed{\frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}}
 \end{array}$$

relator, fusion twice reduce of maps $\cdot \exists \dashv \frac{1}{\exists}$

the last two rows at $X := (R)$:

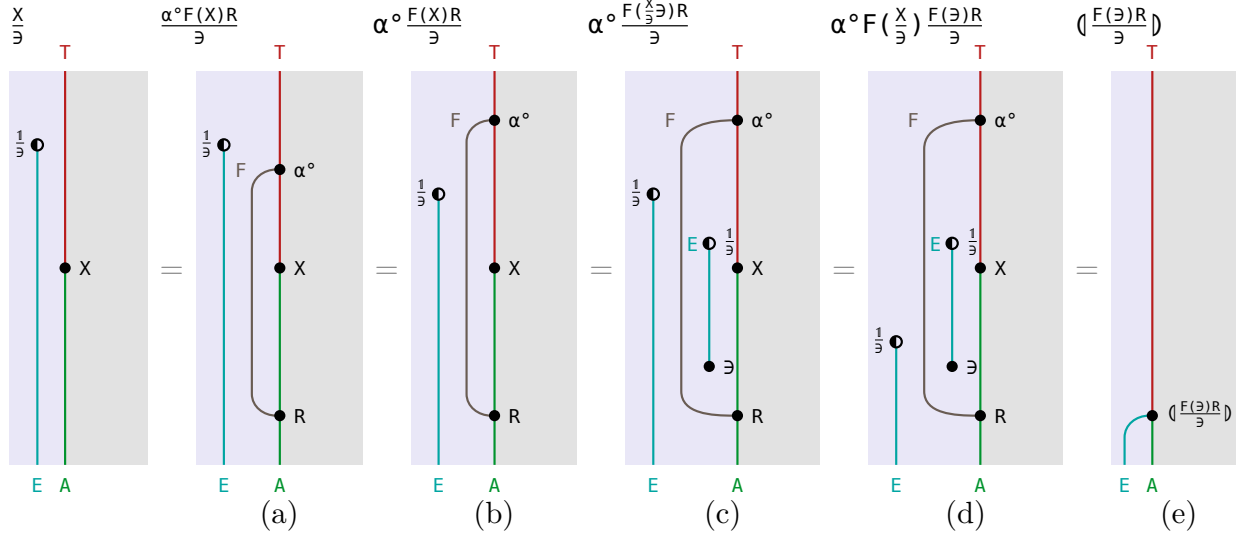
$$\alpha X = F(X)R \iff X = \langle R \rangle$$

(11.6.2c)

in a tabular allegory, with F a relator — so $F(R^\circ) = F(R)^\circ$ — and $\alpha : F(T) \rightarrow T$ its initial algebra, which every later fold theorem assumes: a relation X out of T satisfies the fold equation of the relation R exactly when it is $\langle R \rangle$, the fold of the map $\frac{F(\exists)R}{\exists}$ followed by $\exists$

$$\alpha X = F(X)R \implies \frac{X}{\exists} = \langle \frac{F(\exists)R}{\exists} \rangle$$

a relation X satisfying the fold equation $\alpha X = F(X)R$ of R makes $\frac{X}{\exists}$ the fold of the map $\frac{F(\exists)R}{\exists}$



(a) α iso; $\alpha X = F(X)R$

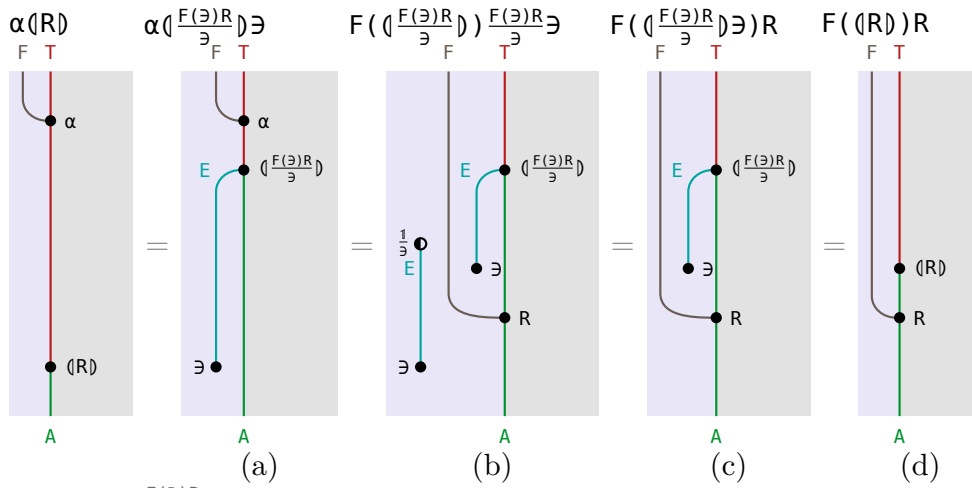
(b) α° is a map because α is an iso, so it leaves Λ : $\frac{\alpha^\circ F(X)R}{\exists} = \alpha^\circ \frac{F(X)R}{\exists}$ — pow-laws

(c) $X = \frac{X}{\exists} \exists$ — pow-laws

(d) $F(\frac{X}{\exists}) = F(\frac{X}{\exists})F(\exists)$ as F is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangle$ relator, pow-laws

(e) fold uniqueness, $\frac{F(\exists)R}{\exists}$ being a map — (11.4a)

$$\alpha \langle R \rangle = F(\langle R \rangle)R$$

 $\Leftarrow$


(a) $\langle R \rangle = \langle \frac{F(\exists)R}{\exists} \rangle \exists$ — (11.6.2b)

(b) the fold's own equation $\alpha(f) = F(\langle f \rangle)f$ at the map $f = \frac{F(\exists)R}{\exists}$ — (11.4a)

(c) $\frac{F(\exists)R}{\exists} \exists = F(\exists)R$ — pow-laws

F a functor — $\triangle$ relator

(d) $\langle \frac{F(\exists)R}{\exists} \rangle \exists = \langle R \rangle$ — (11.6.2b)

§11.6.3. $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$

(11.6.3a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(11.6.3b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 11.6.4 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

§11.6.4. $(S)^\circ (R) = (\mu X : S^\circ F(X)R)$

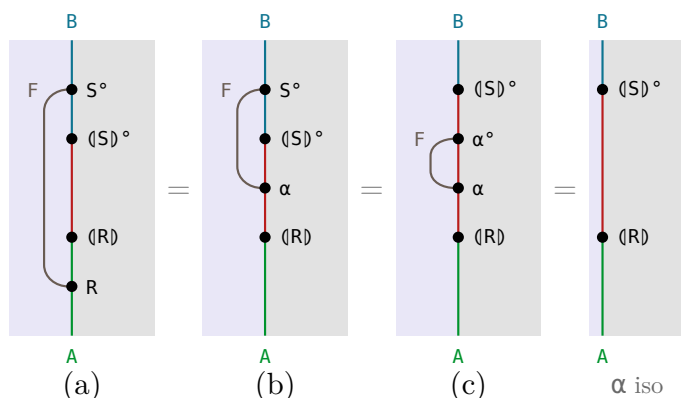
(11.6.4a)

$$(S)^\circ (R) = (\mu X : S^\circ F(X)R)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((S)^\circ (R))R = (S)^\circ (R)$$

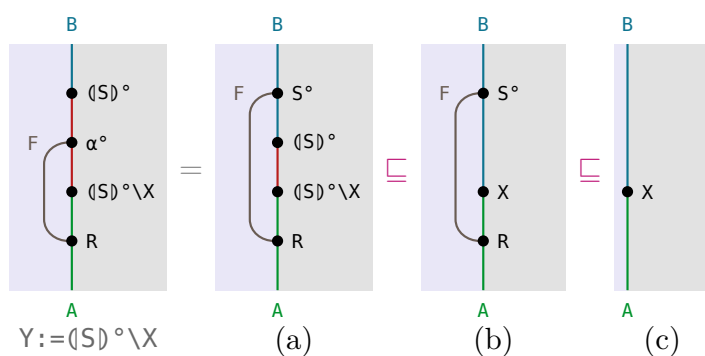
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((S)^\circ (R))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



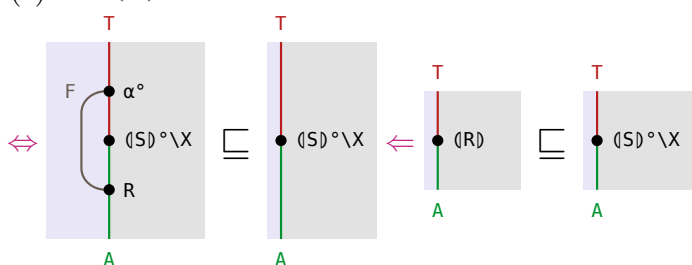
- (a) the body at $(S)^\circ (R)$
 $F(RS) = F(R)F(S) \triangleq \text{relator}$
 (b) $F((R))R = \alpha(R) \triangleq \text{reduce}$
 (c) $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ) \triangleq \text{reduce, (11b)}$

$$S^\circ F(X)R \sqsubseteq X \Rightarrow (S)^\circ (R) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(S)^\circ (R)$ is included in X if X satisfies the associated recursion inequation



- $Y := (S)^\circ \setminus X$
 (a) $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ)$
 $F(RS) = F(R)F(S) \triangleq \text{relator}$
 (b) $(S)^\circ ((S)^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$
 (c) $S^\circ F(X)R \sqsubseteq X$



- $S \cdot \setminus S \triangleq \text{adj-all}$
 (a) $(R) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (11.6.3b), adj-all}$